



Intelligent Decision Support for Evaluating Digital Technologies in Logistics Workforce Management: A CRITIC–TODIM Framework



Vukašin Pajić^{*}, Milan Andrejić[®]

Faculty of Transport and Traffic Engineering, University of Belgrade, 11000 Belgrade, Serbia

^{*} Correspondence: Vukašin Pajić (v.pajic@sf.bg.ac.rs)

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Abstract: The digital transformation of logistics is increasingly driven by interconnected technologies that support workforce coordination, data-driven decision-making, operational integration, and human-centered industrial management. However, logistics companies face difficulties in selecting digital technologies because their adoption involves simultaneous technological, organizational, economic, and employee-related considerations. This study develops an intelligent multi-criteria decision-support framework for evaluating and prioritizing digital technologies for workforce management in logistics companies. Five technology alternatives—artificial intelligence (AI)-based human resources (HR) systems, Internet of Things (IoT) and wearable technologies, enterprise resource planning (ERP)/HR information system (HRIS), digital learning and training, and digital workforce management—are assessed against eight criteria covering implementation complexity, employee productivity, employee satisfaction and retention, implementation and operating costs, scalability and flexibility, data security and privacy risks, legal and ethical risks, and integration capability. The criteria importance through intercriteria correlation (CRITIC) method is applied to derive objective criterion weights from the decision matrix, while interactive and multicriteria decision making (TODIM) is used to rank the alternatives by incorporating the asymmetric effects of gains and losses in the decision process. The results show that Digital Workforce Management achieves the highest overall dominance value (1.000), followed by Digital Learning and Training (0.967) and ERP/HRIS (0.921) systems, whereas AI-based HR systems and IoT and Wearable Technologies rank fourth and fifth, respectively. Robustness analysis using alternative weighting schemes and the technique for order preference by similarity to ideal solution (TOPSIS) and weighted aggregated sum product assessment (WASPAS) methods confirms a relatively stable separation between the three higher-ranked and two lower-ranked alternatives, although the relative positions of the leading technologies vary across methodological settings. The findings demonstrate that digital technology selection in logistics should be treated as an industrial intelligence problem in which operational integration, workforce performance, organizational adaptability, and human-centered risks are evaluated jointly. The proposed framework provides structured decision support for logistics organizations seeking to align digital workforce technologies with broader Industry 4.0 and Industry 5.0 transformation objectives.

Keywords: Industrial intelligence; Intelligent decision support; Logistics digitalization; Digital workforce management; Industry 4.0; Criteria importance through intercriteria correlation; Interactive and multicriteria decision making

1 Introduction

The logistics sector is undergoing an accelerated digital transformation as organizations increasingly integrate artificial intelligence (AI), Internet of Things (IoT) technologies, enterprise information systems, digital learning platforms, and data-driven workforce management tools into their operational environments. These technologies are no longer used solely to automate isolated activities; they increasingly support interconnected decision-making across transportation, warehousing, workforce planning, resource allocation, and organizational coordination. The COVID-19 pandemic further accelerated this transition by increasing the need for resilient, digitally enabled, and remotely manageable logistics processes [1, 2]. As a result, digital transformation has become an important

component of contemporary industrial intelligence, particularly in sectors in which human resources (HR), physical operations, and information systems are tightly interconnected.

Within logistics companies, workforce-related digital technologies play an increasingly important role in supporting operational continuity and organizational responsiveness. AI-based HR systems can assist recruitment, workforce forecasting, performance analysis, and employee development; IoT and wearable technologies can provide real-time information on employee activities, safety, and task execution; enterprise resource planning (ERP) and HR information system (HRIS) platforms enable the integration of workforce information with broader organizational systems; digital learning technologies support continuous competency development; and workforce management systems facilitate scheduling, task allocation, and resource coordination. These applications demonstrate that human resource management (HRM) is becoming increasingly embedded within the broader digital architecture of logistics operations rather than remaining an isolated administrative function [3–5].

However, selecting an appropriate digital technology is not a straightforward technological choice. Different alternatives involve substantially different levels of implementation complexity, expected productivity impact, employee acceptance, investment and operating costs, scalability, integration requirements, data security exposure, and legal or ethical risk. A technology that performs well from an operational perspective may simultaneously create challenges related to privacy, organizational adaptation, or workforce acceptance. Conversely, solutions with lower technological complexity may provide greater compatibility with existing processes and infrastructure. Consequently, technology selection in logistics requires a structured decision process capable of integrating technological, organizational, economic, and human-related dimensions within a unified analytical framework.

This challenge is particularly relevant from the perspective of industrial intelligence. Intelligent industrial decision-making increasingly requires organizations to evaluate heterogeneous data, competing objectives, and conflicting criteria before implementing new technologies. In logistics environments, such decisions are further complicated by the interaction between digital systems and human operators, since the effectiveness of a technology depends not only on its technical capabilities but also on its compatibility with organizational processes, workforce competencies, and existing digital infrastructure. Therefore, evaluating digital workforce technologies can be regarded as a multi-criteria industrial decision problem in which both technological performance and socio-organizational consequences must be considered simultaneously.

Multi-criteria decision-making (MCDM) methods provide an appropriate analytical basis for addressing such problems because they allow multiple alternatives to be evaluated against criteria with different meanings and preference directions. In the present study, the criteria importance through intercriteria correlation and interactive and multicriteria decision making (CRITIC–TODIM) methods are integrated to construct a decision-support framework for the prioritization of digital technologies in logistics workforce management. CRITIC is employed to determine objective criterion weights by considering both the variability of criterion values and the degree of conflict among criteria [6]. TODIM is subsequently used to establish the ranking of the alternatives based on prospect theory, allowing the decision process to reflect the asymmetric perception of gains and losses [7, 8]. The integration of these two methods provides a structured mechanism for combining information-based weighting with behaviorally informed alternative evaluation.

The study evaluates five digital technology alternatives: AI-based HR systems, IoT and Wearable Technologies, ERP/HRIS, Digital Learning and Training, and Digital Workforce Management. These alternatives are assessed according to eight criteria covering implementation complexity, employee productivity impact, employee satisfaction and retention, implementation and operating costs, scalability and flexibility, data security and privacy risks, legal and ethical risks, and integration capability. By jointly considering these dimensions, the proposed framework moves beyond technology selection based solely on cost or operational performance and instead reflects the broader requirements of human-centered and digitally integrated logistics systems.

The contribution of this study is threefold. First, it conceptualizes the selection of digital workforce technologies as an industrial intelligence problem in which technological, organizational, and human factors are jointly evaluated. Second, it develops an integrated CRITIC–TODIM framework for prioritizing alternative technologies in the logistics sector, thereby combining objective criterion weighting with a prospect-theory-based ranking mechanism. Third, the study examines the robustness of the resulting rankings through alternative weighting schemes and comparative MCDM methods, providing additional evidence regarding the stability of the decision outcomes. In this way, the study contributes to the development of structured decision support for logistics organizations seeking to align digital technology investments with operational requirements, workforce needs, and broader Industry 4.0 and Industry 5.0 transformation objectives.

The remainder of the paper is organized as follows. Section 2 reviews the literature on digital technologies, industrial intelligence, workforce management, and technology evaluation in logistics, and identifies the research gap addressed in this study. Section 3 presents the research framework, evaluation criteria, and the CRITIC and TODIM methods. Section 4 reports the application of the proposed framework and the resulting technology rankings. Section 5 presents the robustness analysis and discusses the theoretical and managerial implications of the findings. Finally,

Section 6 summarizes the main conclusions, limitations, and directions for future research.

2 Literature Review and Research Framework

2.1 Digital Transformation and Industrial Intelligence in Logistics

Digitalization and digital transformation are closely related but conceptually distinct. Digitalization generally refers to the incorporation of digital technologies into existing processes, whereas digital transformation represents a broader organizational process in which technologies reshape business models, operating structures, decision mechanisms, and value-creation activities [9]. From this perspective, digital transformation extends beyond the adoption of individual technological tools and involves the systematic integration of digital technologies into organizational processes and managerial practices.

This distinction is particularly important in logistics. Contemporary logistics systems depend on the continuous coordination of transportation, warehousing, inventory, workforce, customer information, and other operational resources. The isolated introduction of a digital technology may improve a specific process, but the broader transformation of logistics operations requires the integration of multiple digital systems and data sources. Digital supply chains therefore increasingly rely on interconnected technologies and information infrastructures that support real-time data acquisition, process coordination, operational visibility, and data-driven decision-making [10].

The technological foundation of this transformation is closely associated with Industry 4.0. Technologies such as the IoT, cloud computing, Big Data analytics, AI, robotics, autonomous systems, and cyber-physical systems enable higher levels of connectivity and automation across industrial and logistics environments [11]. Their value is increasingly derived not only from their individual capabilities, but also from their ability to operate as interconnected components of larger industrial systems. Data collected from sensors, workforce platforms, warehouse systems, transportation systems, and enterprise applications can be integrated and analyzed to support planning, coordination, and operational decision-making.

These developments provide an important foundation for industrial intelligence in logistics. In the context of this study, industrial intelligence refers to the systematic use of data, intelligent technologies, integrated information systems, and decision-support mechanisms to support the performance and adaptability of industrial operations. In logistics companies, this perspective extends beyond physical processes and includes the management and coordination of HR, since employees interact continuously with digital systems, automated technologies, and information-intensive workflows.

2.2 Human-Centric Digitalization and Workforce Transformation

The increasing adoption of digital technologies also changes the nature of work within logistics organizations. As digital systems perform a growing proportion of routine, repetitive, and data-intensive activities, employees increasingly undertake tasks involving interpretation, coordination, problem-solving, communication, and interaction with intelligent technologies [11, 12]. Consequently, digital transformation affects not only operational efficiency but also workforce structure, competency requirements, and the relationship between employees and technology.

These changes have important implications for HRM. HRM is increasingly required to support competency planning, employee development, technological adaptation, and organizational change rather than focusing exclusively on administrative activities [13, 14]. In logistics environments, where operational and workforce processes are closely linked, the ability to align employee capabilities with digital technologies has become particularly important. Previous research has shown that existing workforce qualifications may not always meet the requirements associated with Industry 4.0, highlighting the need for continuous reskilling and upskilling [14].

The competency requirements of logistics professionals have consequently become more diverse. Contemporary logistics work increasingly requires combinations of logistics knowledge, digital literacy, analytical capability, information-system competence, and interpersonal skills [15]. Recent evidence from the maritime logistics sector further highlights the importance of digital literacy, digital HR development, organizational change readiness, and supporting digital infrastructure in the successful implementation of workforce-related digital technologies [16]. This development suggests that technology implementation and workforce development should not be treated as independent managerial activities. Instead, the adoption of digital technologies can generate new competency requirements while simultaneously changing the mechanisms through which employees are trained, monitored, coordinated, and evaluated.

The growing use of digital technologies also introduces new organizational and human-related risks. Systems based on IoT devices, wearable technologies, activity monitoring, AI-supported decision-making, and digital performance assessment can provide valuable real-time information but may also raise concerns regarding employee privacy, data protection, perceived surveillance, transparency, fairness, and organizational trust. Therefore, the evaluation of digital technologies cannot be based exclusively on expected productivity improvements or cost reductions. Human-related risks and employee acceptance should also be incorporated into the decision process.

2.3 Industry 4.0, Industry 5.0, and Human-Centered Industrial Systems

Industry 4.0 provides the technological context for much of the ongoing digital transformation in logistics. Its emphasis on connectivity, automation, cyber-physical integration, and data-driven operations has substantially increased the role of technologies such as IoT, Big Data, cloud computing, machine learning, autonomous systems, and enterprise platforms in industrial environments [11, 17]. These technologies support more responsive and integrated logistics operations while simultaneously changing the structure of work and the competencies required from employees.

However, technology-centered approaches alone do not fully capture the broader organizational consequences of industrial digitalization. Industry 5.0 expands this perspective by emphasizing human-centricity, sustainability, and resilience as complementary dimensions of industrial development [18]. From this viewpoint, technology should support both organizational performance and the long-term development of employees and industrial systems.

This perspective is particularly relevant to logistics workforce management. AI-based systems can support workforce planning and decision-making; digital platforms can improve employee coordination; IoT and wearable devices can contribute to occupational safety and operational monitoring; and digital learning technologies can support continuous competency development. At the same time, these technologies may introduce new ethical, legal, privacy, and organizational challenges. Their evaluation therefore requires simultaneous consideration of technological performance, integration capability, workforce implications, and organizational adaptability.

The transition from Industry 4.0 toward more human-centered industrial systems reinforces the need for intelligent decision-support mechanisms capable of balancing these competing dimensions. From this perspective, technology selection becomes not merely a procurement decision, but a broader industrial intelligence problem in which digital capabilities must be aligned with organizational processes, workforce requirements, and human-centered principles.

2.4 Digital Technology Selection as a Multi-Criteria Decision Problem

Selecting digital technologies for logistics workforce management is inherently a multi-criteria decision problem. The alternatives considered in such decisions may differ substantially in implementation complexity, expected productivity effects, employee satisfaction and retention, financial requirements, scalability, data-security exposure, legal and ethical risks, and compatibility with existing systems.

These criteria may also conflict with one another. A technology with strong productivity potential may require substantial investment, create implementation difficulties, or increase privacy risks. A highly scalable system may require extensive integration with existing digital infrastructure, while a technically advanced solution may face resistance from employees or require substantial competency development. Consequently, a decision based on a single performance indicator is unlikely to capture the full implications of technology adoption.

MCDM methods offer a structured means of addressing such trade-offs because they enable decision alternatives to be evaluated simultaneously against heterogeneous criteria. Within industrial intelligence, MCDM techniques are particularly relevant when decision-making involves technological, economic, organizational, and human considerations that cannot be represented by a single performance measure.

In the present study, the evaluation problem is addressed through an integrated CRITIC–TODIM framework. CRITIC is used to determine the objective importance of the evaluation criteria based on the information contained in the decision matrix, while TODIM is used to rank the technology alternatives by accounting for differences between gains and losses. The combination of the two methods enables the evaluation process to integrate information-based weighting with behaviorally oriented alternative comparison.

2.5 Research Gap and Conceptual Framework

Although previous studies have extensively examined digital transformation, Industry 4.0 technologies, logistics digitalization, workforce competencies, and human-centered industrial development [9–18], these research streams are often considered separately. However, the reviewed literature provides comparatively limited evidence on structured decision frameworks that simultaneously evaluate alternative digital workforce technologies across technological, organizational, economic, and human-centered dimensions.

This gap is particularly relevant because digital technology adoption in logistics is rarely determined by a single objective. Organizations must balance productivity improvements with implementation feasibility, investment requirements, integration capability, workforce acceptance, data protection, and legal or ethical considerations. Therefore, technology selection requires a framework capable of integrating these dimensions within a unified industrial decision-support process.

To address this gap, the present study develops a multi-criteria framework for evaluating five digital technology alternatives for logistics workforce management. The framework incorporates eight evaluation criteria reflecting implementation requirements, workforce outcomes, economic considerations, scalability, technological integration, and risk-related dimensions. CRITIC is employed to derive criterion weights, while TODIM is used to establish the

relative preference ordering of the alternatives. The resulting rankings are subsequently examined through alternative weighting schemes and comparative MCDM methods to assess the stability of the decision outcomes.

The resulting conceptual framework positions digital workforce technology selection as an industrial intelligence problem involving the interaction of digital systems, organizational requirements, and human factors. In this way, the study connects the technological emphasis of Industry 4.0 with the human-centered orientation of Industry 5.0 and provides a structured basis for evaluating digital technologies in logistics environments.

3 Methodology

3.1 Research Framework

A structured multi-criteria decision-support framework was developed to evaluate and prioritize digital technologies for workforce management in logistics companies. The framework integrates the CRITIC method for objective criterion weighting with the TODIM method for alternative ranking. CRITIC determines the relative importance of the evaluation criteria by considering both their contrast intensity and inter-criterion conflict, whereas TODIM ranks the alternatives by incorporating the asymmetric treatment of gains and losses derived from prospect theory [7, 8, 19, 20].

The analytical procedure consists of five principal stages. First, the decision problem is defined and the relevant literature is reviewed. Second, the digital technology alternatives and evaluation criteria are identified. Third, the alternatives are evaluated and the initial decision matrix is constructed. Fourth, the CRITIC and TODIM methods are applied sequentially to determine criterion weights and rank the alternatives. Finally, robustness analysis is conducted using alternative weighting schemes and comparative MCDM methods, followed by interpretation of the decision outcomes.

The overall analytical procedure is illustrated in Figure 1.

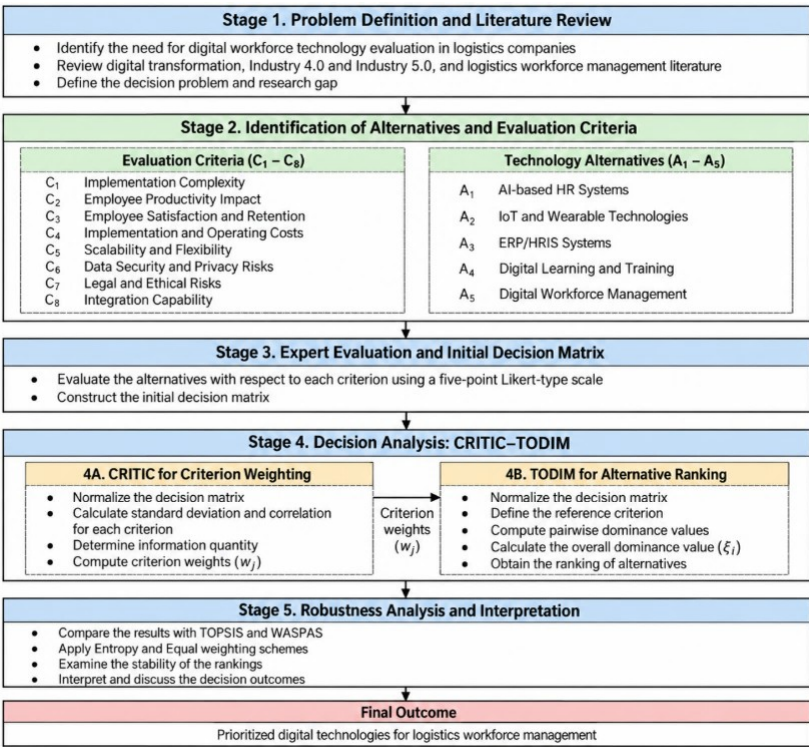


Figure 1. Research framework of the study

Note: AI = artificial intelligence; HR = human resources; IoT = Internet of Things; ERP = enterprise resource planning; HRIS = HR information system; CRITIC = criteria importance through intercriteria correlation; TODIM = interactive and multicriteria decision making; TOPSIS = technique for order preference by similarity to ideal solution; WASPAS = weighted aggregated sum product assessment.

3.2 Digital Technology Alternatives

Five digital technology alternatives were considered in the evaluation: AI-based HR systems (A_1), IoT and Wearable Technologies (A_2), ERP/HRIS systems (A_3), Digital Learning and Training (A_4), and Digital Workforce Management (A_5). These alternatives represent different technological approaches to supporting workforce planning, coordination, development, monitoring, and integration within digitally enabled logistics operations.

AI-Based HR Systems (A1). AI-based HR systems encompass applications that support recruitment and candidate selection, shift planning, employee performance analysis, workforce turnover prediction, and personalized training. Their relevance in logistics is particularly associated with fluctuating labor requirements, complex workforce-allocation problems, and the need for timely personnel decisions. By enabling data-driven decision-making and the automation of selected HR processes, these systems can contribute to more responsive workforce management. However, their implementation also raises challenges related to data privacy, algorithmic transparency, explainability, and potential algorithmic bias.

IoT and Wearable Technologies (A2). IoT-enabled devices, sensors, radio-frequency identification (RFID) technologies, wearable devices, and location-tracking systems can generate real-time data on employee activities, task execution, equipment utilization, and potential ergonomic or safety risks. The information collected through these technologies can support workforce scheduling, performance monitoring, occupational safety, and the identification of employee training needs. Their application can therefore facilitate more data-driven workforce management in operationally intensive logistics environments. Nevertheless, their use may raise concerns regarding employee acceptance, perceived surveillance, privacy, and the protection and appropriate use of employee-related data.

ERP/HRIS Systems (A3). ERP and HRIS systems constitute an important component of the digital infrastructure supporting workforce management. These systems enable the integration and centralization of employee-related information and facilitate the automation of activities such as working-time recording, absence management, payroll administration, and other HR processes. In logistics operations, their integration with Warehouse Management Systems (WMS) and Transportation Management Systems (TMS) is particularly important because it enables workforce information to be connected with operational data and performance indicators.

Digital Learning and Training (A4). Digital learning and training technologies include e-learning platforms, mobile learning solutions, virtual reality applications, and digital systems for monitoring learning progress. In logistics organizations, these technologies can support employee onboarding, occupational safety training, equipment-operation training, technical and interpersonal skill development, and continuous professional development.

Digital Workforce Management (A5). Digital workforce management systems support shift planning, employee availability management, task allocation, absence management, and internal workforce communication. These systems are particularly relevant to logistics environments characterized by fluctuating workloads, shift-based operations, and continuously changing resource requirements. By facilitating data-driven workforce allocation and scheduling, they can improve the alignment between available workforce capacity and operational demand.

3.3 Evaluation Criteria and Initial Decision Matrix

Eight evaluation criteria were defined to capture technological, organizational, economic, workforce-related, and risk-related dimensions associated with digital technology adoption. Recent studies on Industry 5.0, smart logistics, intelligent decision-making, and logistics technology evaluation emphasize the need to consider human-centered, technological, operational, and sustainability-related factors jointly in such decisions [21–25]. This multidimensional structure is also consistent with recent MCDM applications that integrate managerial, technological, and sustainability considerations in complex organizational decisions [26]. The alternatives were evaluated by a panel of experts using a five-point Likert-type scale ranging from 1 to 5.

Criteria C2, C3, C5, and C8 are benefit criteria, for which higher values indicate more desirable performance. In contrast, C1, C4, C6, and C7 are non-benefit criteria, for which lower values are preferable. The definitions, assessment focuses, preference directions, and corresponding evaluation scales of the eight criteria are summarized in Table 1.

Based on the expert assessments, the initial decision matrix was constructed as shown in Table 2.

The general form of the decision matrix is:

$$X = [x_{ij}]_{m \times n}$$

where, m denotes the number of alternatives, n denotes the number of evaluation criteria, and x_{ij} represents the performance of alternative A_i with respect to criterion C_j .

3.4 The Criteria Importance Through Intercriteria Correlation Method

The CRITIC method was applied to determine the objective weights of the evaluation criteria [19, 20]. The method considers both the contrast intensity of each criterion and the degree of conflict between that criterion and the remaining criteria.

Step 1. Normalization of the decision matrix

For benefit criteria,

$$x_{ij}^* = \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}}$$

Table 1. Evaluation criteria

Criterion	Assessment Focus	Scale (1–5)	Preference
C1—Implementation complexity	Time and complexity required for implementation, training requirements, changes to existing processes, technical requirements, vendor support, and potential employee resistance to change	1 = Very low implementation complexity; 3 = Moderate complexity; 5 = Very high implementation complexity	Cost
C2—Employee productivity impact	Impact on work speed and accuracy, task execution, reduction of administrative workload and unproductive time, and workforce utilization	1 = Very small impact; 3 = Moderate impact; 5 = Very large positive impact	Benefit
C3—Employee satisfaction and retention	Impact on employee engagement, motivation, autonomy, professional development, communication, user experience, stress, and intention to remain with the organization	1 = Very unfavorable effect; 3 = Neutral/moderate effect; 5 = Very favorable effect	Benefit
C4—Implementation and operating costs	Acquisition, licensing, infrastructure, integration, training, maintenance, and ongoing operating costs	1 = Very low cost; 3 = Moderate cost; 5 = Very high cost	Cost
C5—Scalability and flexibility	Ability to scale across employees, locations, and HR functions and adapt to different logistics operations and changing organizational requirements	1 = Very limited scalability and flexibility; 3 = Moderate; 5 = Very high scalability and flexibility	Benefit
C6—Data security and privacy risks	Risks related to personal and work-related data, unauthorized access, data storage and transfer, misuse, disclosure, and loss	1 = Very low risk; 3 = Moderate risk; 5 = Very high risk	Cost
C7—Legal and ethical risks	Risks related to regulatory non-compliance, employee rights, transparency, unfairness or discrimination, and accountability	1 = Very low legal and ethical risk; 3 = Moderate risk; 5 = Very high legal and ethical risk	Cost
C8—Integration capability	Ability to integrate with ERP, HRIS, WMS, TMS, payroll, attendance-management systems, and other existing digital infrastructure	1 = Very limited integration capability; 3 = Moderate capability; 5 = Very high integration capability	Benefit

Note: HR = human resources; ERP = enterprise resource planning; HRIS = HR information system; WMS = warehouse management system; TMS = transportation management system.

Table 2. Initial decision matrix

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	5	5	3	5	5	4	5	4
A2	4	5	3	4	4	5	4	4
A3	3	4	4	3	5	3	2	5
A4	2	4	5	3	5	2	2	3
A5	3	5	4	3	5	3	2	5

For non-benefit criteria,

$$x_{ij}^* = \frac{x_j^{\max} - x_{ij}}{x_j^{\max} - x_j^{\min}}$$

Thus,

$$x_{ij}^* = \begin{cases} \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}}, & C_j \text{ is a benefit criterion} \\ \frac{x_j^{\max} - x_{ij}}{x_j^{\max} - x_j^{\min}}, & C_j \text{ is a non-benefit criterion} \end{cases} \quad (1)$$

where,

$$x_j^{\max} = \max_{1 \leq i \leq m} \{x_{ij}\}$$

$$x_j^{\min} = \min_{1 \leq i \leq m} \{x_{ij}\}$$

Following normalization, larger values consistently represent more desirable performance.

Step 2. Determination of criterion information

The mean normalized value of criterion C_j is

$$\bar{x}_j^* = \frac{1}{m} \sum_{i=1}^m x_{ij}^*$$

The standard deviation is calculated as

$$\sigma_j = \sqrt{\frac{\sum_{i=1}^m (x_{ij}^* - \bar{x}_j^*)^2}{m-1}}$$

The Pearson correlation coefficient between criteria C_j and C_k is

$$r_{jk} = \text{corr}(x_{.j}^*, x_{.k}^*).$$

The amount of information contained in criterion C_j is

$$C_j = \sigma_j \sum_{k=1}^n (1 - r_{jk}) \quad (2)$$

where, a larger C_j indicates greater contrast and more non-redundant information.

Step 3. Calculation of objective criterion weights

The objective criterion weight is

$$w_j = \frac{C_j}{\sum_{k=1}^n C_k} \quad (3)$$

The weights satisfy

$$\sum_{j=1}^n w_j = 1, \quad w_j \geq 0$$

3.5 The Interactive and Multicriteria Decision Making Method

TODIM was employed to rank the digital technology alternatives [7, 8]. The method is based on prospect theory and distinguishes explicitly between gains and losses.

Step 1. Construction of the decision matrix

The TODIM decision matrix is

$$S = [s_{ij}]_{m \times n} = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1n} \\ s_{21} & s_{22} & \cdots & s_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ s_{m1} & s_{m2} & \cdots & s_{mn} \end{bmatrix} \quad (4)$$

where, s_{ij} denotes the performance of alternative A_i with respect to criterion C_j .

In the present study,

$$S = X$$

Step 2. Normalization of the decision matrix

For benefit criteria,

$$P_{ij} = \frac{s_{ij}}{\sum_{i=1}^m s_{ij}} \quad (5)$$

For non-benefit criteria,

$$P_{ij} = \frac{\frac{1}{s_{ij}}}{\sum_{i=1}^m \frac{1}{s_{ij}}} \quad (6)$$

For each criterion,

$$\sum_{i=1}^m P_{ij} = 1$$

Step 3. Determination of relative criterion weights

The reference criterion is the criterion with the largest CRITIC weight:

$$w_r = \max_{1 \leq j \leq n} \{w_j\}$$

The relative criterion weight is

$$w_{jr} = \frac{w_j}{w_r} \quad (7)$$

In the present study, criterion C2 has the largest CRITIC weight and therefore serves as the reference criterion. Thus,

$$w_{2r} = 1$$

Step 4. Calculation of pairwise dominance

The dominance of alternative A_i over alternative A_k is

$$\delta(A_i, A_k) = \sum_{j=1}^n \phi_j(A_i, A_k) \quad (8)$$

The partial dominance function is

$$\phi_j(A_i, A_k) = \begin{cases} \sqrt{\frac{w_{jr}(P_{ij} - P_{kj})}{\sum_{l=1}^n w_{lr}}}, & P_{ij} > P_{kj} \\ 0, & P_{ij} = P_{kj} \\ -\frac{1}{\theta} \sqrt{\frac{(P_{kj} - P_{ij}) \sum_{l=1}^n w_{lr}}{w_{jr}}}, & P_{ij} < P_{kj} \end{cases} \quad (9)$$

where, $\theta > 0$ is the loss-attenuation parameter.

For the calculations in this study,

$$\theta = 1$$

Step 5. Calculation of aggregate and overall dominance

The aggregate dominance score of alternative A_i is

$$D_i = \sum_{k=1}^m \delta(A_i, A_k)$$

The minimum aggregate dominance is

$$D_{\min} = \min_{1 \leq q \leq m} \{D_q\}$$

The maximum aggregate dominance is

$$D_{\max} = \max_{1 \leq q \leq m} \{D_q\}$$

The normalized overall dominance value is

$$\xi_i = \frac{D_i - D_{\min}}{D_{\max} - D_{\min}} \quad (10)$$

Equivalently,

$$\xi_i = \frac{\sum_{k=1}^m \delta(A_i, A_k) - \min_{1 \leq q \leq m} \{\sum_{k=1}^m \delta(A_q, A_k)\}}{\max_{1 \leq q \leq m} \{\sum_{k=1}^m \delta(A_q, A_k)\} - \min_{1 \leq q \leq m} \{\sum_{k=1}^m \delta(A_q, A_k)\}}$$

The normalized dominance values satisfy $0 \leq \xi_i \leq 1$.

A larger value of ξ_i indicates stronger overall dominance and therefore a more preferable alternative.

Step 6. Ranking of alternatives

The alternatives are ranked in descending order:

$$\xi_{(1)} > \xi_{(2)} > \cdots > \xi_{(m)}$$

The alternative with the largest overall dominance value is regarded as the most preferable digital technology under the specified evaluation framework.

3.6 Robustness Assessment

The robustness of the CRITIC–TODIM ranking was evaluated using two alternative weighting schemes and two alternative ranking methods. Entropy weighting and equal weighting were combined with TOPSIS and weighted aggregated sum product assessment (WASPAS) to examine whether the principal preference structure remained stable when the weighting and aggregation mechanisms were changed.

The purpose of the robustness assessment is not to require identical rankings under all methodological configurations. Instead, it evaluates whether the broader preference structure remains stable and whether the higher-ranked and lower-ranked groups of alternatives remain distinguishable when methodological assumptions are modified. Particular attention is therefore given to changes in the ordering of the leading technologies and to the stability of the overall separation among the alternatives.

4 Application and Results

The proposed CRITIC–TODIM framework was applied to evaluate the five digital technology alternatives for workforce management in logistics companies. The analysis proceeded in two stages. First, CRITIC was used to determine the objective importance of the eight evaluation criteria. Second, the resulting criterion weights were incorporated into TODIM to determine the dominance and final ranking of the technology alternatives.

4.1 The Criteria Importance Through Intercriteria Correlation-Based Criterion Weighting

The initial decision matrix presented in Table 2 was first normalized according to the preference direction of each criterion using Eq. (1). The resulting normalized decision matrix is presented in Table 3.

Table 3. Normalized decision matrix using the CRITIC method

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	0.000	1.000	0.000	0.000	1.000	0.333	0.000	0.500
A2	0.333	1.000	0.000	0.500	0.000	0.000	0.333	0.500
A3	0.667	0.000	0.500	1.000	1.000	0.667	1.000	1.000
A4	1.000	0.000	1.000	1.000	1.000	1.000	1.000	0.000
A5	0.667	1.000	0.500	1.000	1.000	0.667	1.000	1.000

Note: CRITIC = criteria importance through intercriteria correlation.

The normalized matrix reflects the preference direction of each criterion. Higher normalized values therefore consistently represent more favorable performance. For example, A4 obtains the highest normalized values for implementation complexity, employee satisfaction and retention, implementation and operating costs, data security and privacy risks, and legal and ethical risks. In contrast, A1 and A2 perform particularly strongly in employee productivity impact but less favorably in several implementation- and risk-related dimensions.

The next stage of the CRITIC procedure involved calculating the correlations among the evaluation criteria. The resulting correlation matrix is presented in Table 4.

Table 4. Correlation matrix

	C1	C2	C3	C4	C5	C6	C7	C8
C1	1.000	−0.721	0.943	0.932	0.294	0.808	0.930	−0.157
C2	−0.721	1.000	−0.764	−0.612	−0.408	−0.721	−0.645	0.218
C3	0.943	−0.764	1.000	0.802	0.535	0.943	0.845	−0.286
C4	0.932	−0.612	0.802	1.000	0.250	0.686	0.988	0.200
C5	0.294	−0.408	0.535	0.250	1.000	0.784	0.395	0.134
C6	0.808	−0.721	0.943	0.686	0.784	1.000	0.775	−0.157
C7	0.930	−0.645	0.845	0.988	0.395	0.775	1.000	0.211
C8	−0.157	0.218	−0.286	0.200	0.134	−0.157	0.211	1.000

The correlation structure indicates substantial differences in the information contributed by the criteria. Several implementation- and risk-related criteria are strongly positively correlated. In particular, C4 and C7 exhibit a correlation coefficient of 0.988, while C1 is strongly correlated with C3 and C7. By contrast, C2 exhibits predominantly negative relationships with several other criteria. This comparatively distinct information structure contributes to the higher CRITIC importance subsequently assigned to C2.

The standard deviations of the normalized criteria are presented in Table 5.

Table 5. Standard deviations (SDs) of the normalized criteria

Criterion	C1	C2	C3	C4	C5	C6	C7	C8
SD	0.380	0.548	0.418	0.447	0.447	0.380	0.471	0.418

Among the eight criteria, C2 has the highest standard deviation, with $\sigma_2 = 0.548$ indicating the greatest contrast among the alternatives. C7, C4, and C5 also show relatively high variability, whereas C1 and C6 exhibit the lowest standard deviations.

By combining criterion variability with the degree of conflict among criteria, the CRITIC information quantities shown in Table 6 were obtained.

Table 6. Information quantities obtained using CRITIC

Criterion	C1	C2	C3	C4	C5	C6	C7	C8
C_j	1.509	5.835	1.665	1.679	2.243	1.475	1.650	2.860

Note: CRITIC = criteria importance through intercriteria correlation.

C2 provides the largest information quantity: $C_2 = 5.835$, followed by C8 and C5: $C_8 = 2.860$, $C_5 = 2.243$.

This result indicates that employee productivity impact contributes the greatest amount of distinctive information to the decision problem, while integration capability and scalability and flexibility also play important roles.

The final objective weights derived from the CRITIC method are presented in Table 7.

Table 7. Criterion weights obtained using CRITIC

Criterion	C1	C2	C3	C4	C5	C6	C7	C8
w_j	0.080	0.308	0.088	0.089	0.119	0.078	0.087	0.151

Note: CRITIC = criteria importance through intercriteria correlation.

Employee productivity impact (C2) receives the highest objective weight: $w_2 = 0.308$, followed by integration capability (C8): $w_8 = 0.151$ and scalability and flexibility (C5): $w_5 = 0.119$. Thus,

$$w_2 > w_8 > w_5 > w_4 > w_3 > w_7 > w_1 > w_6$$

The weight assigned to C2 accounts for approximately 30.8% of the total objective criterion importance.

These results indicate that the decision structure is driven primarily by productivity, integration capability, and scalability. However, the comparatively lower weights assigned to risk- and cost-related criteria should not be interpreted as evidence that these dimensions are unimportant. CRITIC weights represent the amount of distinctive information contributed by each criterion within the specific decision matrix rather than an externally imposed judgment of substantive importance.

4.2 The Interactive and Multicriteria Decision Making-Based Evaluation of Digital Technology Alternatives

The CRITIC weights were subsequently incorporated into TODIM to determine the relative preference of the five technology alternatives. Because the initial decision matrix used in TODIM is identical to Table 2, it is not repeated.

Using Eq. (5) and Eq. (6), the values were normalized according to the preference direction of the corresponding criteria. The resulting TODIM normalized decision matrix is presented in Table 8.

Table 8. Normalized decision matrix using the TODIM method

Alternative	C1	C2	C3	C4	C5	C6	C7	C8
A1	0.124	0.217	0.158	0.138	0.208	0.155	0.103	0.190
A2	0.155	0.217	0.158	0.172	0.167	0.124	0.128	0.190
A3	0.206	0.174	0.211	0.230	0.208	0.206	0.256	0.238
A4	0.309	0.174	0.263	0.230	0.208	0.309	0.256	0.143
A5	0.206	0.217	0.211	0.230	0.208	0.206	0.256	0.238

Note: TODIM = interactive and multicriteria decision making.

The normalized values highlight different performance profiles among the alternatives. A4 performs particularly strongly in employee satisfaction and retention and in several dimensions for which lower original complexity or

risk values are preferable. A5 combines high productivity performance with strong integration capability and comparatively favorable implementation and risk characteristics. In contrast, A1 and A2 display strong productivity potential but weaker performance across several implementation, privacy, and legal or ethical dimensions.

Because C2 has the highest CRITIC weight, $w_2 = 0.308$, it was selected as the TODIM reference criterion.

Accordingly, $w_r = w_2 = 0.308$. The relative criterion weights calculated using Eq. (7) are presented in Table 9.

Table 9. Relative criterion weights used in TODIM

Criterion	C1	C2	C3	C4	C5	C6	C7	C8
w_{jr}	0.260	1.000	0.286	0.289	0.386	0.253	0.282	0.490

Note: TODIM = interactive and multicriteria decision making.

For the reference criterion, $w_{2r} = \frac{w_2}{w_2} = 1.000$.

For example, the relative weight of C8 is $w_{8r} = \frac{0.151}{0.308} \approx 0.490$. Similarly, $w_{5r} = \frac{0.119}{0.308} \approx 0.386$. Thus, the relative weighting structure can be expressed as:

$$w_{2r} > w_{8r} > w_{5r} > w_{4r} > w_{3r} > w_{7r} > w_{1r} > w_{6r}$$

Pairwise comparisons were then performed using the TODIM dominance function. With the loss-attenuation parameter specified in Section 3 as $\theta = 1$, the aggregate dominance values presented in Table 10 were obtained.

Table 10. Aggregate dominance values of the alternatives

Alternative	Aggregate Dominance
A1	-18.741
A2	-18.939
A3	-3.017
A4	-2.232
A5	-1.658

Thus, $D_1 = -18.741$, $D_2 = -18.939$, $D_3 = -3.017$, $D_4 = -2.232$, and $D_5 = -1.658$.

The maximum aggregate dominance value is therefore $D_{\max} = D_5 = -1.658$, whereas the minimum is $D_{\min} = D_2 = -18.939$.

The aggregate dominance results reveal a substantial separation between A3–A5 and A1–A2. A5 obtains the most favorable dominance score, followed by A4 and A3. A1 and A2 produce substantially more negative scores.

The aggregate dominance values were then normalized using Eq. (10). For example, the normalized dominance of A5 is

$$\xi_5 = \frac{D_5 - D_{\min}}{D_{\max} - D_{\min}} = \frac{-1.658 - (-18.939)}{-1.658 - (-18.939)} = 1.000$$

For A2,

$$\xi_2 = \frac{-18.939 - (-18.939)}{-1.658 - (-18.939)} = 0.000$$

The resulting overall dominance values and final rankings are presented in Table 11.

Table 11. Overall dominance values and final ranking

Alternative	Overall Dominance	Rank
A1	0.011	4
A2	0.000	5
A3	0.921	3
A4	0.967	2
A5	1.000	1

The normalized dominance values are therefore $\xi_1 = 0.011$, $\xi_2 = 0.000$, $\xi_3 = 0.921$, $\xi_4 = 0.967$, and $\xi_5 = 1.000$.

Hence, the final technology ranking is

$$A5 > A4 > A3 > A1 > A2$$

Or equivalently, Digital Workforce Management > Digital Learning and Training > ERP/HRIS Systems > AI-Based HR Systems > IoT and Wearable Technologies.

The results indicate a clear separation between two groups of alternatives. The higher-performing group is {A3, A4, A5}, whereas the lower-performing group is {A1, A2}.

4.3 Interpretation of the Ranking Results

The first-place ranking of Digital Workforce Management (A5) reflects its balanced performance across the most influential criteria. A5 receives the highest original score for employee productivity impact and strong scores for scalability, integration capability, and several implementation- and risk-related criteria. This combination is particularly influential because the three most important criteria identified by CRITIC are C2, C8, and C5.

Their objective weights are $w_2 = 0.308$, $w_8 = 0.151$, $w_5 = 0.119$.

Together, these three criteria account for $w_2 + w_8 + w_5 = 0.308 + 0.151 + 0.119 = 0.578$.

Thus, approximately 57.8% of the total CRITIC weight is concentrated in employee productivity impact, integration capability, and scalability and flexibility.

Digital Learning and Training (A4), ranked second, performs particularly well in employee satisfaction and retention and exhibits favorable implementation-complexity and risk characteristics. Its weaker integration capability relative to A5 and A3 appears to limit its overall dominance despite its strong performance across several human-centered criteria.

ERP/HRIS systems (A3) rank third and demonstrate a comparatively balanced performance profile. Their strongest characteristics are integration capability and scalability, together with moderate implementation complexity and comparatively favorable legal and ethical risk characteristics. These properties support their role as an important component of integrated digital logistics infrastructure, although their employee-productivity score is lower than those of A1, A2, and A5.

The relative difference between the two leading alternatives is small: $\xi_5 - \xi_4 = 1.000 - 0.967 = 0.033$. Similarly, the difference between A4 and A3 is $\xi_4 - \xi_3 = 0.967 - 0.921 = 0.046$.

By contrast, the difference between A3 and A1 is substantially larger: $\xi_3 - \xi_1 = 0.921 - 0.011 = 0.910$. This confirms that the most pronounced feature of the TODIM results is not the relatively small distinction among A3, A4, and A5, but rather the strong separation between the upper and lower technology groups.

The low rankings of AI-based HR systems (A1) and IoT and Wearable Technologies (A2) should therefore not be interpreted as evidence that these technologies lack potential value. Both alternatives receive the highest original score for employee productivity impact:

$$x_{A1,C2} = x_{A2,C2} = 5$$

However, their overall performance is weakened by less favorable values in implementation complexity, cost, privacy, legal and ethical risk, and related organizational dimensions.

From an industrial intelligence perspective, these findings demonstrate that the most technologically advanced solution is not necessarily the most preferable alternative when implementation feasibility, organizational integration, workforce consequences, scalability, and data-related risks are considered simultaneously. The results therefore support a socio-technical approach to digital technology adoption in which intelligent systems are evaluated according to both their technological capabilities and their compatibility with the broader operational and human environment.

5 Robustness Analysis and Discussion

5.1 Robustness Analysis

To evaluate the stability of the CRITIC–TODIM results, the ranking obtained from the proposed framework was compared with rankings generated using alternative weighting schemes and alternative MCDM methods. Specifically, Entropy weighting and equal weighting were combined with TOPSIS and WASPAS. This design enables the robustness of the results to be examined from two perspectives: sensitivity to the weighting mechanism and sensitivity to the ranking method.

The comparative results are presented in Table 12.

The CRITIC–TODIM ranking obtained in Section 4 was $A5 > A4 > A3 > A1 > A2$.

Under the TOPSIS-Entropy configuration, the ranking becomes $A4 > A5 > A3 > A2 > A1$.

Under TOPSIS with equal weighting, the ranking is $A5 > A3 > A4 > A1 > A2$.

For WASPAS combined with Entropy weighting, the ranking is $A4 > A5 > A3 > A2 > A1$.

Finally, WASPAS with equal weighting produces $A4 > A5 > A3 > A2 > A1$.

These results show that the exact first-place position is not completely invariant across methodological configurations. Digital Workforce Management (A5) ranks first under the CRITIC–TODIM model and TOPSIS

with equal weights, whereas Digital Learning and Training (A4) occupies the first position under TOPSIS-Entropy, WASPAS-Entropy, and WASPAS-Equal.

Table 12. Robustness analysis results

Alternative	TOPSIS-Entropy Score	Rank	TOPSIS-Equal Score	Rank	WASPAS-Entropy Score	Rank	WASPAS-Equal Score	Rank
A1	0.149	5	0.415	4	0.495	5	0.643	5
A2	0.303	4	0.385	5	0.543	4	0.655	4
A3	0.806	3	0.651	2	0.849	3	0.861	3
A4	0.864	1	0.634	3	0.965	1	0.919	1
A5	0.811	2	0.783	1	0.853	2	0.885	2

Note: TOPSIS = technique for order preference by similarity to ideal solution; WASPAS = weighted aggregated sum product assessment.

The variation between A4 and A5 is therefore an important feature of the results rather than a methodological weakness. It indicates that both alternatives exhibit strong overall performance, while their relative preference depends partly on the weighting and aggregation assumptions employed.

A more stable pattern emerges when the alternatives are considered as broader performance groups. Across all tested configurations, A3, A4, and A5 remain within the upper part of the ranking, whereas A1 and A2 consistently occupy the lower positions. This grouping is preserved across all five ranking configurations considered in the study.

The stability of A3 is particularly notable. Under CRITIC–TODIM, TOPSIS-Entropy, WASPAS-Entropy, and WASPAS-Equal, A3 ranks third, while under TOPSIS-Equal it moves to second place. Thus, its rank varies only within the upper-performing group.

The rankings of A4 and A5 exhibit greater variation at the top of the ordering. Their positions across the five configurations can be summarized as $R(A4)=\{2,1,3,1,1\}$ and $R(A5)=\{1,2,1,2,2\}$, where the sequence corresponds to CRITIC–TODIM, TOPSIS-Entropy, TOPSIS-Equal, WASPAS-Entropy, and WASPAS-Equal, respectively.

By contrast, A1 and A2 never enter the top three under any configuration. Their positions remain restricted to fourth and fifth places: $R(A1)=\{4,5,4,5,5\}$, $R(A2)=\{5,4,5,4,4\}$.

The robustness analysis therefore supports two conclusions. First, the broader separation between the higher-performing and lower-performing technology groups is highly stable. Second, the distinction between the leading alternatives, particularly A4 and A5, is sensitive to the selected weighting and ranking approach.

Accordingly, the results should not be interpreted as establishing an absolute superiority of A5 under all possible methodological assumptions. Instead, they indicate that A4 and A5 constitute the two strongest alternatives, while A3 also demonstrates consistently favorable performance.

5.2 Discussion of the Main Findings

The results provide several insights into digital technology selection for workforce management in logistics environments. A primary finding is that technologies offering a balanced combination of workforce coordination, organizational integration, scalability, and manageable implementation requirements perform more favorably than alternatives characterized primarily by advanced technical capabilities.

Digital Workforce Management (A5) achieves the highest ranking in the main CRITIC–TODIM analysis. This result can be explained by its strong performance across several of the most influential criteria. In the initial decision matrix, A5 receives the maximum score for employee productivity impact, scalability and flexibility, and integration capability: $x_{A5,C2} = 5$, $x_{A5,C5} = 5$, $x_{A5,C8} = 5$.

These criteria are also among the most influential according to CRITIC, with weights of $w_2 = 0.308$, $w_5 = 0.119$, and $w_8 = 0.151$.

Their combined contribution is therefore $w_2 + w_5 + w_8 = 0.578$.

This means that 57.8% of the total CRITIC weight is concentrated in productivity, scalability, and integration capability. The strong performance of A5 across these dimensions helps explain its first-place position in the main model.

Digital Learning and Training (A4) performs particularly strongly in human-centered and risk-related dimensions. It receives the highest score for employee satisfaction and retention and favorable scores for implementation complexity, data-security and privacy risk, and legal and ethical risk. This profile is consistent with the human-centric orientation associated with Industry 5.0, in which technological adoption is evaluated not only according to productivity but also according to employee development, adaptability, and organizational resilience.

The robustness analysis further shows that A4 becomes the highest-ranked technology under several alternative methodological configurations. This finding suggests that the relative superiority of A4 or A5 depends partly on whether the evaluation structure places greater emphasis on their different performance characteristics.

ERP/HRIS systems (A3) also exhibit a stable and comparatively strong position. Their principal strength lies in integration capability, combined with favorable scalability and relatively moderate implementation and risk

characteristics. Because logistics organizations increasingly rely on interconnected operational systems, including WMS and TMS, the ability of ERP/HRIS systems to integrate employee-related data with operational information provides an important source of organizational value.

The relatively low rankings of AI-based HR systems (A1) and IoT and Wearable Technologies (A2) deserve careful interpretation. Both alternatives achieve the maximum score for employee productivity impact: $x_{A1,C2} = x_{A2,C2} = 5$.

However, this productivity advantage is offset by less favorable values for implementation complexity, implementation and operating costs, data-security and privacy risks, and legal and ethical risks.

The results therefore do not indicate that AI or IoT technologies are intrinsically inferior. Instead, they show that, within the present evaluation framework, technological sophistication alone is insufficient to guarantee a high overall ranking. Technologies that introduce stronger implementation requirements or risk exposure may be less preferable when evaluated from a broader organizational and workforce perspective.

5.3 Theoretical Implications

The findings support a socio-technical interpretation of industrial intelligence. In digitally enabled logistics systems, intelligent technologies do not operate independently of organizational structures or human actors. Their value depends on the interaction between technical capabilities, workforce requirements, organizational readiness, and existing digital infrastructure.

The study therefore extends the evaluation of industrial intelligence beyond purely technical performance. The decision framework incorporates productivity, employee satisfaction and retention, implementation requirements, scalability, system integration, financial considerations, privacy, and ethical risks within a common analytical structure.

This multidimensional perspective is particularly relevant to the transition from Industry 4.0 toward Industry 5.0. Industry 4.0 has traditionally emphasized connectivity, automation, data exchange, and intelligent systems, whereas Industry 5.0 places greater emphasis on human-centricity, resilience, and the relationship between technological development and employee needs.

The results demonstrate that these perspectives are complementary rather than contradictory. A technology can contribute strongly to industrial intelligence only when its technical capabilities are aligned with the organizational and human environment in which it is implemented.

The ranking behavior of A4 and A5 illustrates this relationship. A5 performs strongly because of its contribution to workforce coordination, productivity, scalability, and system integration, whereas A4 performs strongly because of its contribution to employee development and comparatively favorable implementation and risk characteristics. Both therefore represent different pathways toward human-centered industrial intelligence.

5.4 Managerial Implications

For logistics managers, the results indicate that digital technology investment decisions should not be based solely on technological novelty or anticipated productivity improvement. A structured evaluation should also consider integration capability, implementation complexity, scalability, employee acceptance, privacy, legal exposure, and long-term workforce development.

The strong performance of Digital Workforce Management suggests that logistics companies may obtain substantial value from technologies that directly connect workforce allocation with operational requirements. In environments characterized by variable workloads, shift-based operations, and rapidly changing resource requirements, digital workforce systems can support more responsive personnel allocation and better coordination between workforce capacity and operational demand.

The strong position of Digital Learning and Training also highlights the importance of workforce development during digital transformation. Technology adoption frequently changes task requirements and creates new competency demands. Investments in digital learning should therefore be regarded not merely as supporting HR activities but as part of the broader digital infrastructure required for industrial transformation.

ERP/HRIS systems remain strategically important because they provide an integration layer connecting workforce information with other organizational systems. Their value may be especially significant when organizations already rely on WMS, TMS, payroll, attendance, or other enterprise information platforms.

The comparatively lower rankings of AI-based HR systems and IoT and Wearable Technologies imply that their adoption should be accompanied by more careful implementation planning. Before deployment, organizations should evaluate issues such as employee acceptance, transparency, data governance, privacy protection, algorithmic accountability, and compatibility with existing digital infrastructure.

The findings also suggest that technology prioritization should remain context-dependent. The robustness analysis shows that the relative ordering of the leading alternatives changes when alternative weighting and ranking methods are applied. Managers should therefore avoid interpreting a single numerical ranking as universally optimal.

Instead, the ranking should be used as structured decision support. If organizational priorities change, the criterion weights and technology rankings should be reassessed accordingly.

5.5 Practical Interpretation for Industrial Intelligence

From an industrial intelligence perspective, the results highlight a distinction between technological intelligence and organizational intelligence.

AI-based systems and IoT technologies represent highly advanced technological capabilities. However, their effectiveness depends on whether organizations possess the infrastructure, governance mechanisms, employee competencies, and institutional readiness required for successful implementation.

By contrast, Digital Workforce Management, Digital Learning and Training, and ERP/HRIS systems may provide more immediate organizational value because they directly support coordination, integration, workforce development, and operational alignment.

The findings therefore suggest that industrial intelligence should not be understood simply as increasing the technological sophistication of an organization. A more comprehensive interpretation is required in which intelligent technologies are selected and integrated according to their capacity to support coordinated, adaptable, and human-centered industrial operations.

Accordingly, technological sophistication alone does not guarantee superior organizational outcomes. Effective industrial intelligence depends on the combined alignment of technological capability, organizational fit, workforce readiness, and system integration.

This interpretation provides a stronger connection between the empirical ranking results and the broader objectives of intelligent, human-centered, and digitally integrated logistics systems.

6 Conclusions

This study developed a structured CRITIC–TODIM decision-support framework for evaluating and prioritizing digital technologies for workforce management in logistics companies. Five technology alternatives—AI-based HR systems, IoT and Wearable Technologies, ERP/HRIS systems, Digital Learning and Training, and Digital Workforce Management—were assessed against eight technological, organizational, economic, workforce-related, and risk-related criteria.

The CRITIC analysis showed that employee productivity impact (C2) received the highest objective weight, followed by integration capability (C8) and scalability and flexibility (C5). These three criteria jointly accounted for $0.308 + 0.151 + 0.119 = 0.578$ of the total criterion weight, indicating that productivity, integration, and adaptability play a particularly important role in the present decision structure.

The TODIM analysis ranked the five alternatives as

$$A_5 > A_4 > A_3 > A_1 > A_2$$

Digital Workforce Management (A5) achieved the highest overall dominance value, $\xi_5 = 1.000$, followed by Digital Learning and Training (A4), $\xi_4 = 0.967$, and ERP/HRIS systems (A3), $\xi_3 = 0.921$.

AI-based HR systems (A1) and IoT and Wearable Technologies (A2) obtained substantially lower overall dominance values, $\xi_1 = 0.011$ and $\xi_2 = 0.000$, respectively.

These findings indicate that the most favorable technologies are those that provide a balanced combination of workforce productivity, system integration, scalability, organizational adaptability, and manageable implementation and risk profiles. The results also show that technological sophistication alone does not necessarily lead to a higher overall preference. AI-based and IoT-based solutions may offer strong productivity potential, but their advantages can be offset by implementation complexity, data-security concerns, privacy exposure, and legal or ethical risks.

The robustness analysis further showed that the exact ordering of the leading alternatives is sensitive to the selected weighting and ranking approach. Digital Workforce Management ranks first under the CRITIC–TODIM model, whereas Digital Learning and Training occupies the first position under several alternative TOPSIS and WASPAS configurations. Nevertheless, the broader preference structure remains stable: A3, A4, and A5 consistently form the higher-performing group, while A1 and A2 remain in the lower-performing group.

The study makes three main contributions. First, it frames digital workforce technology selection as an industrial intelligence problem in which technological, organizational, economic, and human factors must be evaluated jointly. Second, it integrates CRITIC and TODIM to combine objective criterion weighting with prospect-theory-based alternative ranking. Third, it demonstrates that robust technology prioritization requires attention not only to productivity and technical capability, but also to organizational integration, workforce development, implementation feasibility, and human-centered risks.

From a practical perspective, the results suggest that logistics organizations should approach digital technology adoption as a structured technology-selection decision rather than as a simple choice of the most advanced technology. Digital Workforce Management, Digital Learning and Training, and ERP/HRIS systems appear

particularly relevant because they support workforce coordination, competency development, system integration, and operational adaptability. AI-based HR systems and IoT and Wearable Technologies may also provide substantial value, but their adoption requires stronger governance, privacy protection, transparency, and organizational readiness.

Several limitations should be acknowledged. First, the assessment is based on expert judgments using a five-point scale, which may introduce subjectivity into the initial decision matrix. Second, the study evaluates a limited set of five technology alternatives and eight criteria, meaning that other digital technologies or organizational considerations may produce different results. Third, the analysis does not incorporate longitudinal evidence from actual technology implementation, and therefore the rankings represent ex ante decision support rather than ex post performance validation. Fourth, the CRITIC weights are derived from a relatively small set of alternatives, which may affect the stability of the inter-criterion correlation structure.

Future research should therefore extend the framework using empirical data from actual logistics companies and evaluate whether the expected benefits and risks identified in the decision model are observed after implementation. Comparative studies across different company sizes, logistics segments, countries, and institutional environments would also help assess the generalizability of the framework. Further research could additionally incorporate fuzzy, interval-valued, or probabilistic decision models to represent uncertainty in expert assessments more explicitly.

The robustness analysis may also be extended by introducing systematic perturbations of criterion weights and TODIM parameters. In particular, future studies could examine the effect of alternative values of the loss-attenuation parameter θ on the final rankings and evaluate whether the leading alternatives remain stable under broader uncertainty scenarios.

Overall, the findings support a broader view of industrial intelligence in which successful digital transformation depends not only on the adoption of advanced technologies but also on their alignment with organizational processes, workforce requirements, and existing digital infrastructure. Effective technology selection should therefore balance technological capability with organizational fit, workforce readiness, and implementation feasibility.

Author Contributions

Conceptualization, M.A. and V.P.; methodology, M.A. and V.P.; software, M.A. and V.P.; writing—original draft preparation, M.A. and V.P.; writing—review and editing, M.A. and V.P. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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